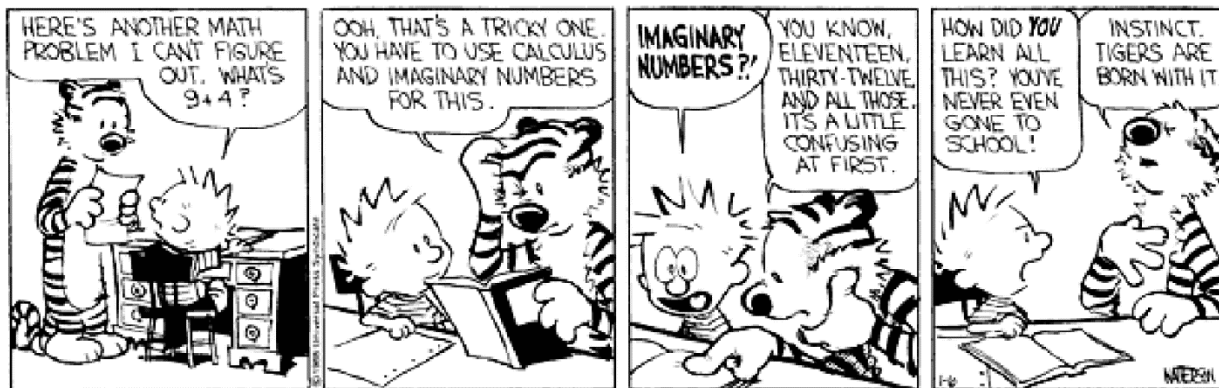




ECEN 314 – Signals and Systems

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Functions of Complex Variables



Learning Objectives



From ECEN 214 you should remember how to:

- ✓ Be able to express complex numbers in Cartesian and polar form.
- ✓ Be able to perform arithmetic on complex numbers.

From these slides:

- ✓ Be able to evaluate elementary functions (polynomials, exponentials, logs, trig functions) of complex numbers.
- ✓ Be able to use complex algebra to simplify the evaluation of certain integrals.
- ✓ Be able to find fractional powers of complex numbers.

Bonus material:

- ✓ Be able to express inverse trig functions in terms of logs.

Complex Numbers – Exponentials and Logs



Exponentials: Evaluating exponentials of complex numbers is easiest to do using the Cartesian form. Euler's identity along with properties of exponents can be used to help us evaluate exponentials of complex numbers.

$$\begin{aligned}e^z &= e^{x+jy} = e^x e^{jy} \\ &= e^x \cos(y) + j e^x \sin(y).\end{aligned}$$

Logs (Natural): Evaluating logs of complex numbers is easiest using the polar form. Here we just use log properties to help us.

$$\begin{aligned}\ln(z) &= \ln(re^{j\theta}) \\ &= \ln(r) + \ln(e^{j\theta}) \\ &= \ln(r) + j\theta.\end{aligned}$$

Note: When dealing with complex numbers, $f(z) = \ln(z)$ is what is called a multi-valued function. Suppose for example we wanted to find $\log(-5)$. The number -5 can be written as

$$-5 = 5e^{j\pi}$$

so that

$$\begin{aligned}\ln(-5) &= \ln(5e^{j\pi}) \\ &= \ln(5) + j\pi.\end{aligned}$$

But we could have just as easily (and correctly) written

$$-5 = 5e^{-j\pi}$$

resulting in

$$\begin{aligned}\ln(-5) &= \ln(5e^{-j\pi}) \\ &= \ln(5) - j\pi.\end{aligned}$$

In fact, $\ln(-5)$ has an infinite number of values:

$$\ln(-5) = \ln(5) - j\pi + j2\pi n$$

for any integer n .

Exponential and Log Examples



Example 1.1

(a) Evaluate e^z if $z = 3 + 4j$.

(b) Evaluate e^z if $z = 5e^{j4\pi/3}$.

(c) Evaluate $\ln(z)$ if $z = 4e^{-j\pi/3}$.

(d) Evaluate $\ln(z)$ if $z = 4 - 3j$.

Complex Exponentials and Trig Functions



- Euler's identity allows us to relate complex exponentials and trig functions.

$$e^{j\theta} = \cos(\theta) + j\sin(\theta) \leftarrow \text{Euler's identity (1)}$$

$$e^{-j\theta} = \cos(\theta) - j\sin(\theta) \leftarrow \text{Conjugate (2)}$$

$$\rightarrow \cos(\theta) = \text{Re}[e^{j\theta}]$$

$$\rightarrow \sin(\theta) = \text{Im}[e^{j\theta}]$$

- Adding (1) and (2) and dividing by 2 gives

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}.$$

- Subtracting (1) and (2) and dividing by $2j$ gives

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}.$$

Note: These relationships between exponentials and trig functions make it quite easy to develop all sorts of trig identities. For example:

$$\cos^2(\theta) = \frac{e^{j2\theta} + 2 + e^{-2j\theta}}{4},$$

$$\sin^2(\theta) = \frac{-e^{j2\theta} + 2 - e^{-2j\theta}}{4}.$$

- ✓ Adding these two expressions proves that $\cos^2(\theta) + \sin^2(\theta) = 1$.

- ✓ Or, from the first expression

$$\begin{aligned} \cos^2(\theta) &= \frac{1}{2} \left[1 + \frac{e^{j2\theta} + e^{-2j\theta}}{2} \right] \\ &= \frac{1}{2} [1 + \cos(2\theta)] \end{aligned}$$

Complex Exponential and Trig Function Identities



- Now that we have written basic trig functions in terms of exponentials, we can evaluate trig functions of complex arguments. For example

$$\begin{aligned}\cos(z) &= \cos(x + jy) = \frac{e^{j(x+jy)} + e^{-j(x+jy)}}{2} = \frac{1}{2}e^{-y}e^{jx} + \frac{1}{2}e^ye^{-jx} \\ &= \frac{1}{2}e^{-y}(\cos(x) + j\sin(x)) + \frac{1}{2}e^y(\cos(x) - j\sin(x)) \\ &= \frac{1}{2}\cos(x)(e^y + e^{-y}) + j\frac{1}{2}\sin(x)(e^{-y} - e^y)\end{aligned}$$

- ✓ $\cos(x + jy) = \cos(x)\cosh(y) - j\sin(x)\sinh(y)$

- Similarly

- ✓ $\sin(x + jy) = \sin(x)\cosh(y) + j\cos(x)\sinh(y)$

- ✓ $\cosh(x + jy) = \cosh(x)\cos(y) + j\sinh(x)\sin(y)$

- ✓ $\sinh(x + jy) = \sinh(x)\cos(y) + jcosh(x)\sin(y)$

Example 1.2



Example 1.2

- a) Evaluate $\sin(z)$ if $z = 3 + 4j$.
- b) Evaluate $\sin(z)$ if $z = \frac{1}{2}e^{-j\pi/2}$.

Integrals Involving Exponentials and Trig Functions



- We can use our knowledge of complex functions to help us to do things in the real world. Suppose, for example, we wanted to evaluate an integral involving exponentials and trig functions such as

$$\int e^{at} \cos(bt) dt$$

- In your Calculus class you were probably taught to compute this integral using integration by parts (twice). But it might be easier to use complex functions as follows

$$\begin{aligned} \cos(bt) &= \operatorname{Re}[e^{jbt}] \\ e^{at} \cos(bt) &= e^{at} \cdot \operatorname{Re}[e^{jbt}] = \operatorname{Re}[e^{at} e^{jbt}] = \operatorname{Re}[e^{(a+jb)t}] \\ \int e^{at} \cos(bt) dt &= \int \operatorname{Re}[e^{(a+jb)t}] dt = \operatorname{Re}\left[\int e^{(a+jb)t} dt\right] = \operatorname{Re}\left[\frac{e^{(a+jb)t}}{a+jb}\right] \end{aligned}$$

- Now we write the complex expression in its Cartesian (rectangular) form to extract the real part

$$\operatorname{Re}\left[\frac{e^{(a+jb)t}}{a+jb}\right] = \operatorname{Re}\left[\frac{e^{(a+jb)t}(a-jb)}{a^2+b^2}\right] = \frac{e^{at}}{a^2+b^2} (a \cdot \cos(bt) + b \cdot \sin(bt))$$

Example 1.3



Example 1.3 - Evaluate $\int e^{-4t} \sin(2\pi t) dt$.

Integrals Involving Exponentials and Trig Functions – Fourier Type Integrals



- In this class, we will also run into integrals similar to the last slide, but slightly different,

$$\int e^{jat} \cos(bt) dt$$

- In this case, the procedure from the last slide won't help (why not?) but a slightly different procedure will. This time, write the cosine as

$$e^{jat} \cos(bt) = e^{jat} \cdot \frac{e^{jbt} + e^{-jbt}}{2} = \frac{e^{j(a+b)t} + e^{j(a-b)t}}{2}$$

$$\int e^{jat} \cos(bt) dt = \frac{1}{2} \int e^{j(a+b)t} dt + \frac{1}{2} \int e^{j(a-b)t} dt$$

$$\begin{aligned} &= \frac{e^{j(a+b)t}}{2j(a+b)} + \frac{e^{j(a-b)t}}{2j(a-b)} = \frac{\cos((a+b)t) + j\sin((a+b)t)}{2j(a+b)} + \frac{\cos((a-b)t) + j\sin((a-b)t)}{2j(a-b)} \\ &= \left[\frac{\sin((a+b)t)}{2(a+b)} + \frac{\sin((a-b)t)}{2(a-b)} \right] - j \left[\frac{\cos((a+b)t)}{2(a+b)} + \frac{\cos((a-b)t)}{2(a-b)} \right] \end{aligned}$$

Note: In this example, the original integrand was complex, so it is not surprising that the result is complex as well.

Example 1.4



Example 1.4 - Evaluate $\int e^{j2t} \sin(3\pi t) dt$.

Roots of Unity



- We can build polynomial functions from powers of a complex variable, z .

$$z^2 = (x + jy)^2 = (x^2 - y^2) + 2jxy \quad \text{or} \quad z^2 = (re^{j\theta})^2 = r^2 e^{j2\theta},$$

$$z^3 = (x + jy)^3 = (x^3 - 3xy^2) + j(3x^2y - y^3) \quad \text{or} \quad z^3 = (re^{j\theta})^3 = r^3 e^{j3\theta},$$

$\vdots$

$$z^n = \sum_{k=0}^n \binom{n}{k} (j)^k x^{n-k} y^k \quad \text{or} \quad z^n = r^n e^{jn\theta} = r^n [\cos(n\theta) + j\sin(n\theta)]$$

- Things get a little more interesting when we look at fractional powers. Suppose we wanted to solve the algebraic equation $z^n - 1 = 0$. $z = (1)^{1/n}$ (known as the n roots of unity).

$$n = 2 \rightarrow z^2 - 1 = (z + 1)(z - 1) = 0 \rightarrow z = \pm 1.$$

$$n = 3 \rightarrow z^3 - 1 = (z^2 + z + 1)(z - 1) = 0 \rightarrow z = 1, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}.$$

$\vdots$

- In general, we could write $1 = e^{j2\pi k}$ for any integer k . Then

$$z = (1)^{1/n} = (e^{j2\pi k})^{1/n} = e^{j2\pi k/n}.$$

- ✓ Distinct roots are obtained for $k = 0, 1, 2, \dots, n - 1$, and then they repeat after that.
- ✓ The n roots of unity turn out to be equally spaced along the unit circle in the complex plane.

(Ir)Rational Powers of Complex Numbers



- Roots of other complex numbers can be written in terms of the roots of unity,

$$z^{1/n} = (re^{j\theta})^{1/n} = (re^{j(\theta+2\pi k)})^{1/n} = r^{1/n} e^{j\theta/n} e^{j2\pi k/n}.$$

There are n distinct roots which are all multiples of the roots of unity.

- We can extend this idea to any rational power,

$$z^{m/n} = (re^{j\theta})^{m/n} = (re^{j(\theta+2\pi k)})^{m/n} = r^{m/n} e^{j\theta m/n} e^{j2\pi km/n}.$$

As long as the integers m, n have no factors in common (relatively prime), then the term $e^{j2\pi km/n}$ will cycle through the n roots of unity (not necessarily in order) as k ranges from 0 to $n - 1$.

- Irrational powers can be handled using the log function. If α is some irrational number, then

$$z^\alpha = e^{\ln(z^\alpha)} = e^{\alpha \ln(z)} = e^{\alpha \ln(re^{j(\theta+2\pi k)})} = e^{\alpha [\ln(r) + j(\theta+2\pi k)]} = r^\alpha e^{j\alpha(\theta+2\pi k)}.$$

Now since α is irrational, the term $2\alpha\pi k$ does not cycle through integer multiples of 2π and there are infinitely many values.

- For all of these multi-valued functions, we refer to the value where $k = 0$ as the *principal value*.

Powers of Complex Numbers



Example 1.5

- a) Determine all possible values of $j^{1/2}$.

- b) Determine all possible values of $(1 - j)^{3/4}$.

Bonus Material: Inverse Trig Functions



- It is common to express inverse trig functions of a complex argument in terms of log functions. Consider $w = \cos^{-1}(z)$. In this case, w is the solution to the equation $\cos(w) = z$. Writing $\cos(z)$ in terms of exponentials

$$\frac{e^{jw} + e^{-jw}}{2} = z.$$

- Make the substitution, $u = e^{jw} \rightarrow \frac{1}{u} = e^{-jw} \rightarrow w = -j\ln(u)$. Then the equation becomes

$$u + \frac{1}{u} = 2z \rightarrow u^2 - 2zu + 1 = 0 \rightarrow u = z + (z^2 - 1)^{1/2}$$

$$w = \cos^{-1}(z) = -j\ln\left(z + (z^2 - 1)^{1/2}\right).$$

Here it is understood that the term $(z^2 - 1)^{1/2}$ is two valued and the $\ln()$ term has infinitely many values. So in fact there are two infinite sets of values for the inverse cosine.

- We could come up with similar expressions for other inverse trig functions, e.g.,

$$\sin^{-1}(z) = -j\ln\left(jz + (1 - z^2)^{1/2}\right),$$

$$\tan^{-1}(z) = \frac{j}{2}\ln\left(\frac{j+z}{j-z}\right).$$